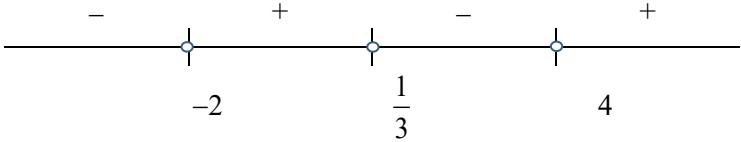
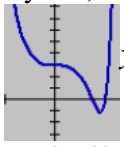


Qn	Solution	Comments
1	$\frac{4x^2 + 4x - 14}{x - 4} - (x + 3) = \frac{4x^2 + 4x - 14 - (x + 3)(x - 4)}{x - 4}$ $= \frac{4x^2 + 4x - 14 - x^2 + x + 12}{x - 4}$ $= \frac{3x^2 + 5x - 2}{x - 4}$ $\frac{4x^2 + 4x - 14}{x - 4} < x + 3$ $\frac{4x^2 + 4x - 14}{x - 4} - (x + 3) < 0$ $\frac{3x^2 + 5x - 2}{x - 4} < 0$ $\frac{(3x - 1)(x + 2)}{x - 4} < 0$  $x < -2 \text{ or } \frac{1}{3} < x < 4$	
2i	By GC, gradient of curve at $x = 0$ is 0 Gradient of curve at $x = \frac{\pi}{2}$ is -0.693	We can use the nDeriv function of the GC. (Press ALPHA WINDOW 3) Make sure angle is in radians.
ii	At $x = 0, y = 2$ Equation of tangent at $x = 0$ is $y = 2$. At $x = \frac{\pi}{2}, y = 1$ $y - 1 = -0.69315 \left(x - \frac{\pi}{2} \right)$ $y = -0.69315x + 2.0888$ Equation of tangent at $x = \frac{\pi}{2}$ is $y = -0.693x + 2.09$ Solving simultaneous equation, $2 = -0.69315x + 2.0888$ $x = 0.12810$ The point of intersection is $(0.128, 2)$.	

Qn	Solution	Comments
3	$y = k(x-l)^4 + m \Rightarrow \frac{dy}{dx} = 4k(x-l)^3$ $0 = 4k(a-l)^3$ Since $c > b$, $k \neq 0$ (Read Comments). $\therefore a = l$ $b = k(a-l)^4 + m$ $= m$ $c = k(l)^4 + m$ $= ka^4 + b$ $k = \frac{c-b}{a^4}$ Alternative Read Comments column first. Under the sequence of transformation, the turning point of $y = x^4$ is mapped onto the turning point of $y = f(x)$. $(0,0) \xrightarrow{A} (l,0) \xrightarrow{B} (l,0) \xrightarrow{C} (l,m)$, where A: translation of l units in the positive x-direction B: stretching by factor k, parallel to the y-axis, x-axis invariant C: translation of m units in the positive y-direction Hence $l = a$, $m = b$. Substitute $x = 0$, $y = c$, $c = k(0-a)^4 + b \Rightarrow k = \frac{c-b}{a^4}$ Since $c > b$, $k > 0$.	If $k = 0$, $c = b = m$ Alternative To start this question, we need to know the sequence of transformation involved in this question $y = x^4 \rightarrow y = (x-l)^4$ Translate the graph l units in the positive x-direction $y = (x-l)^4 \rightarrow y = k(x-l)^4$ Stretch the resultant graph by factor k parallel to the y-axis, x-axis invariant $y = k(x-l)^4 \rightarrow y = k(x-l)^4 + m$ Translate the resultant graph m units in the positive y-direction. Your graph should show that the values of a, b and c are positive with $c > b$.

Qn	Solution	Comments
4i	$a + 3d = br^4 \quad \text{--- (1)}$ $a + 8d = br^7 \quad \text{--- (2)}$ $a + 11d = br^{14} \quad \text{--- (3)}$ $(2) - (1): 5d = br^7 - br^4 \Rightarrow d = \frac{br^4(r^3 - 1)}{5}$ $(3) - (2): 3d = br^{14} - br^7 \Rightarrow d = \frac{br^7(r^7 - 1)}{3}$ $\frac{br^4(r^3 - 1)}{5} = \frac{br^7(r^7 - 1)}{3}$ $\frac{br^4}{15}(3(r^3 - 1) - 5r^3(r^7 - 1)) = 0$ Since b and r are non-zero, $\frac{br^4}{15} \neq 0$ $3(r^3 - 1) - 5r^3(r^7 - 1) = 0$ $5r^{10} - 5r^3 - 3r^3 + 3 = 0$ $5r^{10} - 8r^3 + 3 = 0$ By GC,  $y = 5x^{10} - 8x^3 + 3$ $r = 0.740$	We want to form an equation in r, so we need to remove a, b and d in the process. Since it is given that $r < 1$, we can solve the equation using a graph, and looking the region where $x < 1$. Can also use PlySmlt2, but it takes a long time and have to scroll through many complex roots.
ii	Method 1: Required sum = $S_\infty - S_n$ $= \frac{b}{1-r} - \frac{b(1-r^n)}{1-r}$ $= \frac{br^n}{1-r}$ $= 3.8528b(0.74045)^n$ $\approx 3.85b(0.740)^n$ Method 2: Required sum = $\frac{br^n}{(n+1)^{\text{th}} \text{ term}} + \frac{br^{n+1}}{(n+2)^{\text{th}} \text{ term}} + \frac{br^{n+2}}{(n+3)^{\text{th}} \text{ term}} + \dots$ $= \frac{br^n}{1-r}$ $= 3.8528b(0.74045)^n$ $\approx 3.85b(0.740)^n$	Use at least 5dp for value of r in your calculations

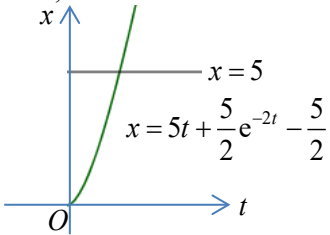
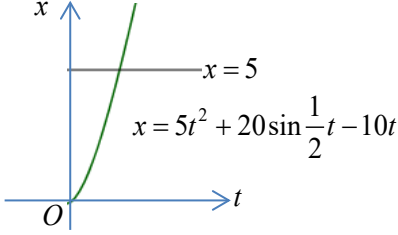
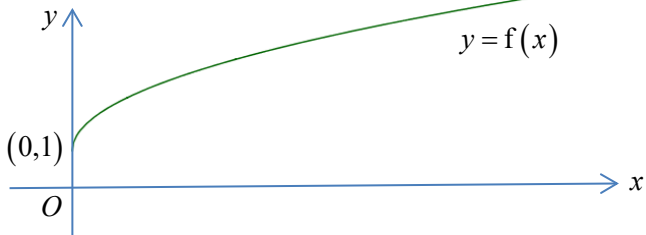
Qn	Solution	Comments
5i	Method 1: $(\underline{u} + \underline{v}) \times (\underline{u} - \underline{v}) = \underline{u} \times \underline{u} - \underline{u} \times \underline{v} + \underline{v} \times \underline{u} - \underline{v} \times \underline{v}$ Since $\underline{u} \times \underline{u} = \underline{v} \times \underline{v} = \underline{0}$ and $-\underline{u} \times \underline{v} = \underline{v} \times \underline{u}$, $(\underline{u} + \underline{v}) \times (\underline{u} - \underline{v}) = \underline{0} + \underline{v} \times \underline{u} + \underline{v} \times \underline{u} - \underline{0}$ $= 2\underline{v} \times \underline{u}$ $= 2 \begin{pmatrix} a \\ 0 \\ b \end{pmatrix} \times \begin{pmatrix} 2 \\ -1 \\ 2 \end{pmatrix}$ $= 2 \begin{pmatrix} b \\ 2b - 2a \\ -a \end{pmatrix}$ Method 2: $(\underline{u} + \underline{v}) \times (\underline{u} - \underline{v}) = \begin{pmatrix} 2+a \\ -1 \\ 2+b \end{pmatrix} \times \begin{pmatrix} 2-a \\ -1 \\ 2-b \end{pmatrix}$ $= \begin{pmatrix} b - 2 + 2 + b \\ -((2+a)(2-b) - (2+b)(2-a)) \\ -2 - a + 2 - a \end{pmatrix}$ $= \begin{pmatrix} 2b \\ -(4 + 2a - 2b - ab - 4 - 2b + 2a + ab) \\ -2a \end{pmatrix}$ $= \begin{pmatrix} 2b \\ 4b - 4a \\ -2a \end{pmatrix}$	
ii	If the $\underline{i}$- and $\underline{k}$- components are equal, $b = -a$ $(\underline{u} + \underline{v}) \times (\underline{u} - \underline{v}) = \begin{pmatrix} -2a \\ -4a - 4a \\ -2a \end{pmatrix} = \begin{pmatrix} -2a \\ -8a \\ -2a \end{pmatrix}$ If $(\underline{u} + \underline{v}) \times (\underline{u} - \underline{v})$ is a unit vector, $\left \begin{pmatrix} -2a \\ -8a \\ -2a \end{pmatrix} \right = 1$ $\sqrt{4a^2 + 64a^2 + 4a^2} = 1$ $\sqrt{72a^2} = 1$ $ a = \frac{1}{6\sqrt{2}}$ $a = \pm \frac{1}{6\sqrt{2}}$	Recall that $\sqrt{a^2} = a $.

Qn	Solution	Comments
iii	$(u + v) \cdot (u - v) = u \cdot u - u \cdot v + v \cdot u - v \cdot v$ $0 = u ^2 - v ^2$ $ v ^2 = u ^2$ $= \sqrt{4 + 1 + 4}$ $= 9$ $ v = 3 \text{ (Since } v > 0 \text{)}$	
6i	Let P_n be the statement $\sum_{r=1}^n r(r^2 + 1) = \frac{1}{4}n(n+1)(n^2 + n + 2)$, for all positive integers n. Check P_1: LHS = $\sum_{r=1}^1 r(r^2 + 1) = 1(1^2 + 1) = 2$ RHS = $\frac{1}{4}1(1+1)(1^2 + 1 + 2) = 2$ $\therefore P_1$ is true. Assume that P_k is true for some positive integer k i.e. $\sum_{r=1}^k r(r^2 + 1) = \frac{1}{4}k(k+1)(k^2 + k + 2)$. Want to show that P_{k+1} is true i.e. $\sum_{r=1}^{k+1} r(r^2 + 1) = \frac{1}{4}(k+1)(k+2)((k+1)^2 + (k+1) + 2)$ $= \frac{1}{4}(k+1)(k+2)(k^2 + 2k + 1 + k + 1 + 2)$ $= \frac{1}{4}(k+1)(k+2)(k^2 + 3k + 4)$ LHS of $P_{k+1} = \sum_{r=1}^{k+1} r(r^2 + 1)$ $= \frac{1}{4}k(k+1)(k^2 + k + 2) + (k+1)((k+1)^2 + 1)$ $= \frac{1}{4}(k+1)(k^3 + k^2 + 2k + 4(k^2 + 2k + 2))$ $= \frac{1}{4}(k+1)(k^3 + k^2 + 2k + 4k^2 + 8k + 8)$ $= \frac{1}{4}(k+1)(k^3 + 5k^2 + 10k + 8)$ $= \frac{1}{4}(k+1)(k+2)(k^2 + 3k + 4)$ $= \text{RHS of } P_{k+1}$ $\therefore P_k$ is true $\Rightarrow P_{k+1}$ is true	Not in H2 Math syllabus

Qn	Solution	Comments
	Since P_1 is true and P_k is true $\Rightarrow P_{k+1}$ is true, P_n is true for all positive integers n .	
ii	$u_1 = u_0 + n^3 + n = 2 + 1 + 1 = 4$ $u_2 = u_1 + n^3 + n = 4 + 8 + 2 = 14$ $u_3 = u_2 + n^3 + n = 14 + 27 + 3 = 44$	
iii	$u_r = u_{r-1} + r^3 + r$ $u_r - u_{r-1} = r^3 + r$ $\sum_{r=1}^n (u_r - u_{r-1}) = \sum_{r=1}^n (r^3 + r)$ $\begin{array}{r} \cancel{u_1} - u_0 \\ + \cancel{u_2} - \cancel{u_1} \\ \quad \vdots \\ + \cancel{u_{n-1}} - \cancel{u_{n-2}} \\ + u_n - \cancel{u_{n-1}} \end{array} = \frac{1}{4}n(n+1)(n^2 + n + 2)$ $u_n - u_0 = \frac{1}{4}n(n+1)(n^2 + n + 2)$ $u_n = \frac{1}{4}n(n+1)(n^2 + n + 2) + 2$	$\sum_{r=1}^n (u_r - u_{r-1})$ hints at method of difference using the recurrence relation given in (ii). We will also need to see the link between different parts in the entire question. We will need to use the result from (i) to simplify RHS.
7a	Method 1: $w^2 + (-1 - 8i)w + (-17 + 7i)$ $= (-1 + 5i)^2 + (-1 - 8i)(-1 + 5i) + (-17 + 7i)$ $= 1 - 10i - 25 + 1 + 3i + 40 - 17 + 7i$ $= 0$ Hence, $-1 + 5i$ is a root of the equation, Let z be the second root $w^2 + (-1 - 8i)w + (-17 + 7i) = (w + 1 - 5i)(w - z)$ Coefficient of w: $(1 - 5i) - z = -1 - 8i$ $z = 2 + 3i$ The second root is $2 + 3i$ Method 2: $\begin{array}{r} w + (-2 - 3i) \\ w + 1 - 5i \overline{) w^2 + (-1 - 8i)w + (-17 + 7i)} \\ - (w^2 + (1 - 5i)w) \\ \hline (-2 - 3i)w + (-17 + 7i) \\ - ((-2 - 3i)w + (-17 + 7i)) \\ \hline 0 \end{array}$ Since the remainder is 0, $-1 + 5i$ is a root of the equation, and the second root is $2 + 3i$.	Note that this is a question that does not allow us to use the GC. Hence all working needs to be shown clearly. It is not advisable to use the quadratic formula for this question, because we will need to show the working for evaluating $\sqrt{b^2 - 4ac}$. However, we can still use the quadratic formula to check answers.

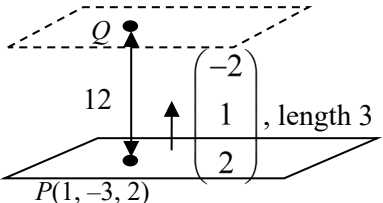
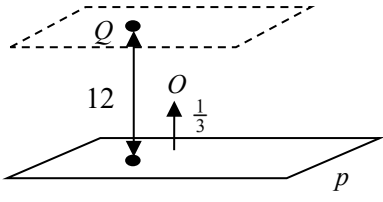
Qn	Solution	Comments
b	Method 1: $z^3 - 5z^2 + 16z + k = 0$ $(1 + ai)^3 - 5(1 + ai)^2 + 16(1 + ai) + k = 0$ $1 + 3ai - 3a^2 - a^3i - 5 - 10ai + 5a^2 + 16 + 16ai + k = 0$ $(12 + k + 2a^2) + ai(9 - a^2) = 0$ Comparing imaginary part, $a(9 - a^2) = 0$ Since a is a positive real constant, $a = 3$. Comparing real part, $12 + k + 2(3)^2 = 0 \Rightarrow k = -30$ Method 2: Since k is a real value, by conjugate root theorem, $1 - ai$ is also a root. $(z - (1 + ai))(z - (1 - ai)) = z^2 - 2z + 1 + a^2$ Let the third root be w. $z^3 - 5z^2 + 16z + k = (z^2 - 2z + 1 + a^2)(z - w)$ Coefficient of z^2: $-w - 2 = -5 \Rightarrow w = 3$ Coefficient of z: $2w + 1 + a^2 = 16$ $a^2 = 9 \Rightarrow a = 3$ (Since $a > 0$) Constant term: $k = -3(1 + 3^2) \Rightarrow k = -30$	To expand z^3, we can use binomial expansion. (Given in MF 26) Be careful with signs and bracketing when using this method. We can use the polysmlt app to verify our answers.
8i	$f(x) = \tan(ax + b)$ $f'(x) = a \sec^2(ax + b)$ $= a(1 + \tan^2(ax + b))$ $= a(1 + y^2)$ $f''(x) = a \left(2y \frac{dy}{dx} \right)$ $= 2ay(a(1 + y^2))$ $= 2a^2(y + y^3)$ $f'''(x) = 2a^2 \frac{dy}{dx} (1 + 3y^2)$ $= 2a^3(1 + y^2)(1 + 3y^2)$	Read the instructions carefully, we need to express the answers in terms of a and y.
ii	When $b = \frac{\pi}{4}$, $f(0) = 1$ $f'(0) = 2a, f''(0) = 4a^2, f'''(0) = 2a^3(2)(4) = 16a^3$	

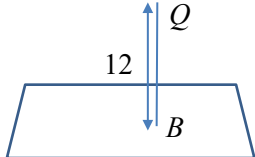
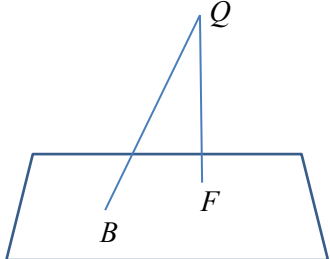
Qn	Solution	Comments
	$f(x) = 1 + 2ax + 2a^2x^2 + \frac{16a^3}{3!}x^3 + \dots$ $= 1 + 2ax + 2a^2x^2 + \frac{8a^3}{3}x^3 + \dots$	
iii	When $a = 2$ and $b = 0$, $f(0) = 0, f'(0) = 2, f''(0) = 0, f'''(0) = 16$ $\tan(2x) = 2x + \frac{16}{3!}x^3 + \dots$ $= 2x + \frac{8}{3}x^3 + \dots$	
9ia	$y = \frac{dx}{dt} \Rightarrow \frac{dy}{dt} = \frac{d^2x}{dt^2}$ $\frac{d^2x}{dt^2} + 2\frac{dx}{dt} = 10$ $\frac{dy}{dt} + 2y = 10$ $\frac{dy}{dt} = 10 - 2y$	This is a show question. $\frac{dy}{dt} = \frac{d^2x}{dt^2}$ must be seen in the working.
b	$\frac{dy}{dt} = 10 - 2y$ $\int \frac{1}{10 - 2y} dy = \int 1 dt$ $-\frac{1}{2} \ln 10 - 2y = t + C$ $\ln 10 - 2y = -2t - 2C$ $ 10 - 2y = Ae^{-2t}$ $10 - 2y = Be^{-2t}$ $y = 5 - \frac{B}{2}e^{-2t}$ When $t = 0$, $\frac{dx}{dt} = y = 0$, $B = 10$ $y = 5 - 5e^{-2t}$ $\frac{dx}{dt} = 5 - 5e^{-2t}$ $x = 5t + \frac{5}{2}e^{-2t} + D$ When $t = 0$, $x = 0$, $D = -\frac{5}{2}$ $x = 5t + \frac{5}{2}e^{-2t} - \frac{5}{2}$	Need to show the removal of modulus properly. Alternative Method Since the initial y (or $\frac{dx}{dt}$) is 0, $10 - 2y > 0.$ $-\frac{1}{2} \ln(10 - 2y) = t + C$

Qn	Solution	Comments
ii	$\frac{d^2x}{dt^2} = 10 - 5\sin\frac{1}{2}t$ $\frac{dx}{dt} = 10t + 10\cos\frac{1}{2}t + E$ When $t = 0$, $\frac{dx}{dt} = 0$, $E = -10$ $\frac{dx}{dt} = 10t + 10\cos\frac{1}{2}t - 10$ $x = 5t^2 + 20\sin\frac{1}{2}t - 10t + F$ When $t = 0$, $x = 0$, $F = 0$ $x = 5t^2 + 20\sin\frac{1}{2}t - 10t$	
iii	For model 1, $5 = 5t + \frac{5}{2}e^{-2t} - \frac{5}{2}$ By GC,  $t = 1.4738 \approx 1.47$ For model 2, $5 = 5t^2 + 20\sin\frac{1}{2}t - 10t$  $t = 1.0460 \approx 1.05$	
10ai	Let $y = 1 + \sqrt{x}$ $x = (y-1)^2$ $f^{-1}(x) = (x-1)^2$  $D_{f^{-1}} = R_f = [1, \infty)$	

Qn	Solution	Comments
ii	$ff(x) = x$ $1 + \sqrt{1 + \sqrt{x}} = x$ $\sqrt{1 + \sqrt{x}} = x - 1$ $1 + \sqrt{x} = x^2 - 2x + 1$ $\sqrt{x} = x^2 - 2x$ $x = x^4 - 4x^3 + 4x^2$ $x(x^3 - 4x^2 + 4x - 1) = 0$ Since $1 + \sqrt{1 + \sqrt{0}} = 2 \neq 0$, $x \neq 0$ $x^3 - 4x^2 + 4x - 1 = 0$ By GC, $x = 0.38197, 1, 2.6180$ $1 + \sqrt{1 + \sqrt{0.38197}} = 2.2720 \neq 0.38197$ $1 + \sqrt{1 + \sqrt{1}} = 2.4142 \neq 1$ $1 + \sqrt{1 + \sqrt{2.6180}} = 2.6180$ Hence, $x = 2.62$ (3sf) Since $2.62 \in D_{f^{-1}}$, $f^{-1}ff(x) = f^{-1}(x)$ $f(x) = f^{-1}(x)$	We do not just divide by x throughout the equation. We factorize and check that $x = 0$ is not a solution. There are a few extra solutions in this question. They came about when we square both sides of the equation. Hence, we have to check the solutions against the initial equation.
bi	$g(4) = 2 + g(2)$ $= 4 + g(1)$ $= 5 + g(0)$ $= 6$ $g(7) = 1 + g(6)$ $= 3 + g(3)$ $= 4 + g(2)$ $= 8$ $g(12) = 2 + g(6)$ $= 9$	
ii	$g(8) = 2 + g(4)$ $= 8$ Since $g(7) = g(8)$, g is not a one-one function, g^{-1} does not exist.	For this portion, we are looking for different values of x that give the same value of g . This is similar to the idea of finding $g(a) = g(b)$ for 2 different values of a and b . It involves a bit of trial and error.
11ia	Method 1: $\begin{pmatrix} 1 \\ 2 \\ 0 \end{pmatrix} \cdot \begin{pmatrix} -2 \\ 1 \\ 2 \end{pmatrix} = -2 + 2 + 0 = 0 \quad \text{and} \quad \begin{pmatrix} 0 \\ 4 \\ -2 \end{pmatrix} \cdot \begin{pmatrix} -2 \\ 1 \\ 2 \end{pmatrix} = 0 + 4 - 4 = 0$	

Qn	Solution	Comments
	Since l is perpendicular to 2 vectors parallel to the plane p, l is perpendicular to p Method 2: $\begin{pmatrix} 1 \\ 2 \\ 0 \end{pmatrix} \times \begin{pmatrix} 0 \\ 4 \\ -2 \end{pmatrix} = \begin{pmatrix} -4 \\ 2 \\ 4 \end{pmatrix} = 2 \begin{pmatrix} -2 \\ 1 \\ 2 \end{pmatrix}$ Since normal vector of p is parallel to direction vector of l, l is perpendicular to p. Method 1: $\begin{pmatrix} 1 \\ -3 \\ 2 \end{pmatrix} + \lambda \begin{pmatrix} 1 \\ 2 \\ 0 \end{pmatrix} + \mu \begin{pmatrix} 0 \\ 4 \\ -2 \end{pmatrix} = \begin{pmatrix} -1 \\ 0 \\ 1 \end{pmatrix} + t \begin{pmatrix} -2 \\ 1 \\ 2 \end{pmatrix}$ $\begin{pmatrix} \lambda & + & 2t \\ 2\lambda & + & 4\mu & - & t \\ - & 2\mu & - & 2t \end{pmatrix} = \begin{pmatrix} -2 \\ 3 \\ -1 \end{pmatrix}$ By GC, $\lambda = -\frac{8}{9}$, $\mu = \frac{19}{18}$, $t = -\frac{5}{9}$ Method 2: $\begin{pmatrix} 1 \\ -3 \\ 2 \end{pmatrix} \cdot \begin{pmatrix} -2 \\ 1 \\ 2 \end{pmatrix} = -1$ Equation of p is $\vec{r} \cdot \begin{pmatrix} -2 \\ 1 \\ 2 \end{pmatrix} = -1$ $\left(\begin{pmatrix} -1 \\ 0 \\ 1 \end{pmatrix} + t \begin{pmatrix} -2 \\ 1 \\ 2 \end{pmatrix} \right) \cdot \begin{pmatrix} -2 \\ 1 \\ 2 \end{pmatrix} = -1$ $4 + 9t = -1$ $t = -\frac{5}{9}$ $\begin{pmatrix} 1 \\ -3 \\ 2 \end{pmatrix} + \lambda \begin{pmatrix} 1 \\ 2 \\ 0 \end{pmatrix} + \mu \begin{pmatrix} 0 \\ 4 \\ -2 \end{pmatrix} = \begin{pmatrix} -1 \\ 0 \\ 1 \end{pmatrix} - \frac{5}{9} \begin{pmatrix} -2 \\ 1 \\ 2 \end{pmatrix} = \begin{pmatrix} \frac{1}{9} \\ -\frac{5}{9} \\ -\frac{1}{9} \end{pmatrix}$ $\vec{i}$-component: $1 + \lambda = \frac{1}{9} \Rightarrow \lambda = -\frac{8}{9}$ $\vec{k}$-component: $2 - 2\mu = -\frac{1}{9} \Rightarrow \mu = \frac{19}{18}$	

Qn	Solution	Comments
b	(Read the comments column first if you are not sure how to start) Method 1: A vector perpendicular to p is $\begin{pmatrix} -2 \\ 1 \\ 2 \end{pmatrix}$ which has magnitude 3. $\therefore$ vectors of length 12 perpendicular to $p =$ $\pm \frac{12}{3} \begin{pmatrix} -2 \\ 1 \\ 2 \end{pmatrix} = \pm \begin{pmatrix} -8 \\ 4 \\ 8 \end{pmatrix}$ Since $P(1, -3, 2)$ is on p, then letting Q be a point on the plane that has perpendicular distance 12 from p (see diagram on the right), $\overrightarrow{OQ} = \overrightarrow{OP} + \overrightarrow{PQ} = \begin{pmatrix} 1 \\ -3 \\ 2 \end{pmatrix} \pm \begin{pmatrix} -8 \\ 4 \\ 8 \end{pmatrix} = \begin{pmatrix} -7 \\ 1 \\ 10 \end{pmatrix} \text{ or } \begin{pmatrix} 9 \\ -7 \\ -6 \end{pmatrix}$ $\therefore$ the two planes have equations $\underline{r} \cdot \begin{pmatrix} -2 \\ 1 \\ 2 \end{pmatrix} = \begin{pmatrix} -7 \\ 1 \\ 10 \end{pmatrix} \cdot \begin{pmatrix} -2 \\ 1 \\ 2 \end{pmatrix} \quad \underline{r} \cdot \begin{pmatrix} -2 \\ 1 \\ 2 \end{pmatrix} = \begin{pmatrix} 9 \\ -7 \\ -6 \end{pmatrix} \cdot \begin{pmatrix} -2 \\ 1 \\ 2 \end{pmatrix}$ ie. $-2x + y + 2z = 35$ and $-2x + y + 2z = -37$ Method 2: Recall: Equation of p is $\underline{r} \cdot \begin{pmatrix} -2 \\ 1 \\ 2 \end{pmatrix} = -1$ The shortest distance from O to $p = \frac{ -1 }{\sqrt{(-2)^2 + 1^2 + 2^2}} = \frac{1}{3}$ Equation of plane that is $12 - \frac{1}{3} = \frac{35}{3}$ units away from O is $\underline{r} \cdot \begin{pmatrix} -2 \\ 1 \\ 2 \end{pmatrix} = 3 \left(\frac{35}{3} \right) \text{ i.e. } \underline{r} \cdot \begin{pmatrix} -2 \\ 1 \\ 2 \end{pmatrix} = 35$ Equation of plane that is $12 + \frac{1}{3} = \frac{37}{3}$ units away from O is $\underline{r} \cdot \begin{pmatrix} -2 \\ 1 \\ 2 \end{pmatrix} = 3 \left(-\frac{37}{3} \right) \text{ i.e. } \underline{r} \cdot \begin{pmatrix} -2 \\ 1 \\ 2 \end{pmatrix} = -37$	 

Qn	Solution	Comments
	Method 3: Let m be a line perpendicular to p, Q be the point on m that is 12 units from p and B be $(1, -3, 2)$ One of the equations of m is $\vec{r} = \begin{pmatrix} 1 \\ -3 \\ 2 \end{pmatrix} + s \begin{pmatrix} -2 \\ 1 \\ 2 \end{pmatrix}, s \in \mathbb{R}$. $\vec{OQ} = \begin{pmatrix} 1 \\ -3 \\ 2 \end{pmatrix} + s \begin{pmatrix} -2 \\ 1 \\ 2 \end{pmatrix}$ for some value of s. $\vec{BQ} = \begin{pmatrix} 1 \\ -3 \\ 2 \end{pmatrix} + s \begin{pmatrix} -2 \\ 1 \\ 2 \end{pmatrix} - \begin{pmatrix} 1 \\ -3 \\ 2 \end{pmatrix} = s \begin{pmatrix} -2 \\ 1 \\ 2 \end{pmatrix}$ $BQ = \left s \begin{pmatrix} -2 \\ 1 \\ 2 \end{pmatrix} \right = 3 s = 12$ $s = 4 \Rightarrow s = \pm 4$ $\vec{OQ} = \begin{pmatrix} 1 \\ -3 \\ 2 \end{pmatrix} \pm 4 \begin{pmatrix} -2 \\ 1 \\ 2 \end{pmatrix} = \begin{pmatrix} -7 \\ 1 \\ 10 \end{pmatrix}$ or $\begin{pmatrix} 9 \\ -7 \\ -6 \end{pmatrix}$ $\begin{pmatrix} -7 \\ 1 \\ 10 \end{pmatrix} \cdot \begin{pmatrix} -2 \\ 1 \\ 2 \end{pmatrix} = 35$ or $\begin{pmatrix} 9 \\ -7 \\ -6 \end{pmatrix} \cdot \begin{pmatrix} -2 \\ 1 \\ 2 \end{pmatrix} = -37$ Equation of the planes are $-2x + y + 2z = 35$ and $-2x + y + 2z = -37$ Method 4: Let Q be the point on l such that Q is 12 units from p, point F be $\left(\frac{1}{9}, -\frac{5}{9}, -\frac{1}{9}\right)$ and point B be $(1, -3, 2)$. $\vec{OQ} = \begin{pmatrix} -1 \\ 0 \\ 1 \end{pmatrix} + s \begin{pmatrix} -2 \\ 1 \\ 2 \end{pmatrix}$, for some value of s. $\vec{BQ} = \begin{pmatrix} -1 \\ 0 \\ 1 \end{pmatrix} + s \begin{pmatrix} -2 \\ 1 \\ 2 \end{pmatrix} - \begin{pmatrix} 1 \\ -3 \\ 2 \end{pmatrix} = \begin{pmatrix} -2 \\ 3 \\ -1 \end{pmatrix} + s \begin{pmatrix} -2 \\ 1 \\ 2 \end{pmatrix}$ $FQ = \left \left(\begin{pmatrix} -2 \\ 3 \\ -1 \end{pmatrix} + s \begin{pmatrix} -2 \\ 1 \\ 2 \end{pmatrix} \right) \cdot \frac{1}{3} \begin{pmatrix} -2 \\ 1 \\ 2 \end{pmatrix} \right$	For this part, the other planes that we are looking for are parallel to p, else the planes will intersect. Our aim is to find points Q which are 12 units from p (or point B, which is in p). We will use line m to help us find Q. (Refer to diagram below)  We can also use $\left(\frac{1}{9}, -\frac{5}{9}, -\frac{1}{9}\right)$ as point B, because it is in p. Be careful when removing the modulus, there are 2 possible answers for s. For Method 2, we are also finding point Q. However, we are using the method of length of projection. 

Qn	Solution	Comments
	$\frac{1}{3} 5+9s =12$ $ 5+9s =36$ $5+9s=\pm 36$ $s=\frac{31}{9} \text{ or } -\frac{41}{9}$ $\overrightarrow{OQ}=\begin{pmatrix} -1 \\ 0 \\ 1 \end{pmatrix}+\frac{31}{9}\begin{pmatrix} -2 \\ 1 \\ 2 \end{pmatrix}=\begin{pmatrix} -\frac{71}{9} \\ \frac{31}{9} \\ \frac{71}{9} \end{pmatrix} \text{ or }$ $\overrightarrow{OQ}=\begin{pmatrix} -1 \\ 0 \\ 1 \end{pmatrix}-\frac{41}{9}\begin{pmatrix} -2 \\ 1 \\ 2 \end{pmatrix}=\begin{pmatrix} \frac{73}{9} \\ -\frac{41}{9} \\ -\frac{73}{9} \end{pmatrix}$ $\begin{pmatrix} -\frac{71}{9} \\ \frac{31}{9} \\ \frac{71}{9} \end{pmatrix} \cdot \begin{pmatrix} -2 \\ 1 \\ 2 \end{pmatrix}=35 \text{ or } \begin{pmatrix} \frac{73}{9} \\ -\frac{41}{9} \\ -\frac{73}{9} \end{pmatrix} \cdot \begin{pmatrix} -2 \\ 1 \\ 2 \end{pmatrix}=-37$ Equation of the planes are $-2x+y+2z=35$ and $-2x+y+2z=-37$.	Be careful when removing the modulus, there are 2 possible answers for s.
ii	$\vec{n}=\begin{pmatrix} 1 \\ 2 \\ 0 \end{pmatrix} \times \begin{pmatrix} a \\ 4 \\ -2 \end{pmatrix}=\begin{pmatrix} -4 \\ 2 \\ 4-2a \end{pmatrix}$ Since l is parallel to p, $\begin{pmatrix} -4 \\ 2 \\ 4-2a \end{pmatrix} \cdot \begin{pmatrix} -2 \\ 1 \\ 2 \end{pmatrix}=0$ $18-4a=0 \Rightarrow a=\frac{9}{2}$	If l and p do not meet at a unique point, l is parallel to p.